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```
clc;
clear;
close all;
```

# CALCULATION OF PROCESSING GAIN AND GENERATION OF PN SEQUENCE

Spread Spectrum Communication

## Processing Gain Parameters

```
Rb = 1000;           % Data Rate (bits/second)
Rc = 10000;          % Chip Rate (chips/second)

% Calculate Processing Gain
PG = Rc / Rb;

% Calculate Processing Gain in dB
PG_dB = 10 * log10(PG);

% Display Processing Gain
fprintf('=====\\n');
fprintf('      SPREAD SPECTRUM COMMUNICATION\\n');
fprintf('=====\\n');

fprintf('Data Rate      = %d bps\\n', Rb);
fprintf('Chip Rate       = %d chips/s\\n', Rc);
fprintf('Spreading Factor = %.2f\\n', PG);
fprintf('Processing Gain = %.2f\\n', PG);
fprintf('Processing Gain = %.2f dB\\n', PG_dB);

=====
      SPREAD SPECTRUM COMMUNICATION
=====
Data Rate      = 1000 bps
Chip Rate       = 10000 chips/s
Spreading Factor = 10.00
```

---

```
Processing Gain = 10.00
Processing Gain = 10.00 dB
```

## Generate PN Sequence

```
% Length of PN sequence
pn_length = 31;

% Generate random binary PN sequence
pn = randi([0 1], 1, pn_length);

% Display PN Sequence
fprintf('\n=====\\n');
fprintf('          PN SEQUENCE\\n');
fprintf('=====\\n');

disp('Generated PN Sequence:');
disp(pn);
```

```
=====
          PN SEQUENCE
=====
Generated PN Sequence:
Columns 1 through 13

    0    0    0    0    1    1    0    1    0    0    0    1    1

Columns 14 through 26

    0    0    0    1    1    1    0    1    1    0    0    0    1

Columns 27 through 31

    0    1    0    1    0
```

## Count Number of 0s and 1s

```
num_zeros = sum(pn == 0);
num_ones = sum(pn == 1);

fprintf('Number of 0s = %d\\n', num_zeros);
fprintf('Number of 1s = %d\\n', num_ones);

Number of 0s = 18
Number of 1s = 13
```

## Plot PN Sequence

```
figure;
```

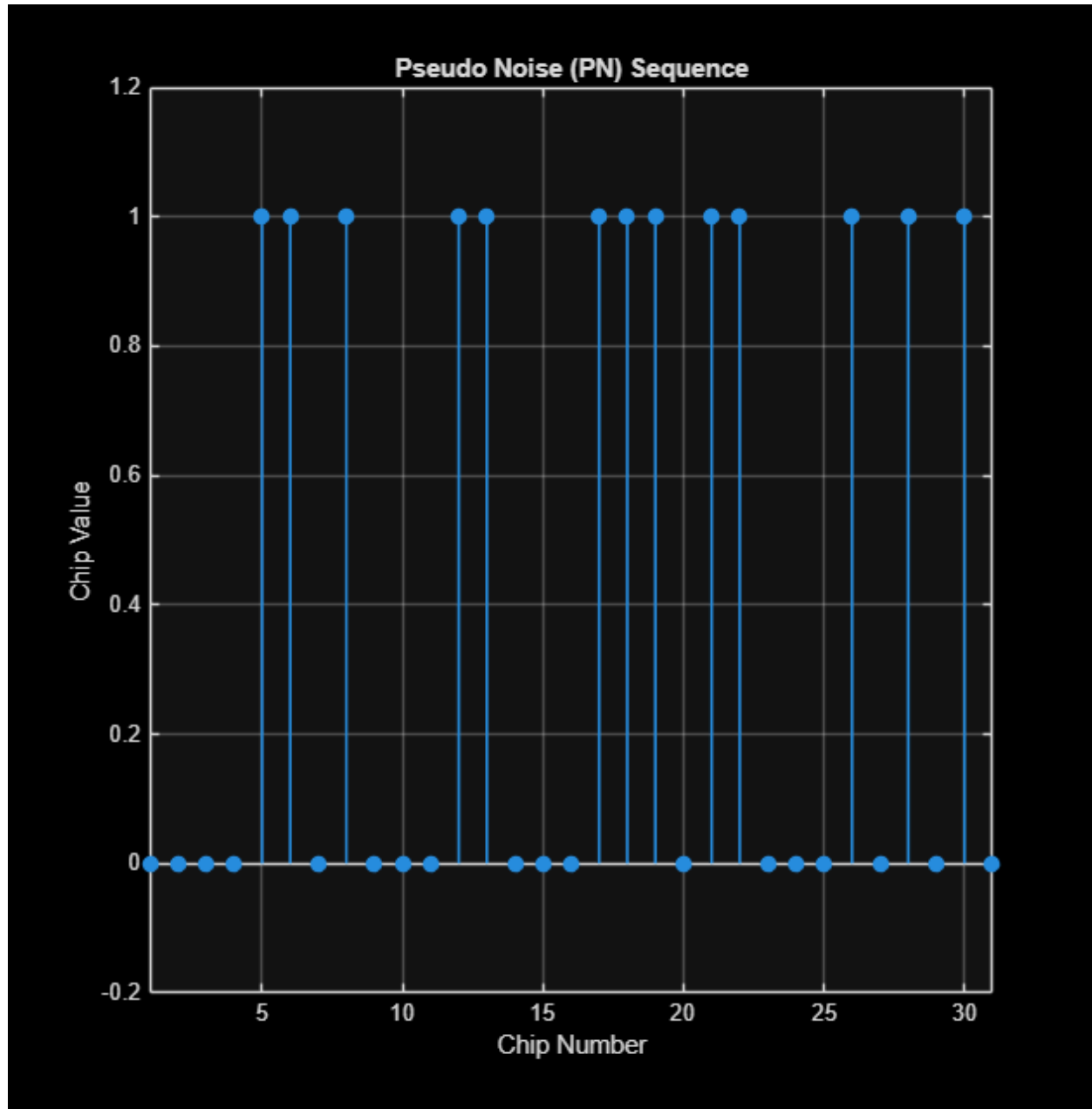
---

```
stem(1:pn_length, pn, 'filled', 'LineWidth', 1.2);

title('Pseudo Noise (PN) Sequence');
xlabel('Chip Number');
ylabel('Chip Value');

ylim([-0.2 1.2]);
xlim([1 pn_length]);

grid on;
```



---

# Display Bandwidth Expansion

```
fprintf('\n=====\\n');
fprintf('          SYSTEM PARAMETERS\\n');
fprintf('=====\\n');

fprintf('Spreading Factor      = %.2f\\n', PG);
fprintf('Bandwidth Expansion    = %.2f times\\n', PG);
fprintf('Processing Gain        = %.2f dB\\n', PG_dB);

fprintf('\\n=====\\n');
fprintf('Program execution completed successfully.\\n');
fprintf('=====\\n');
```

```
=====
          SYSTEM PARAMETERS
=====
Spreading Factor      = 10.00
Bandwidth Expansion    = 10.00 times
Processing Gain        = 10.00 dB

=====
Program execution completed successfully.
=====
```

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